

Gear Contact Geometry

A typed finite observation system

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Status. This document supersedes the bilingual working draft v1.0. It is migrated to GO Core v0.2 and the Frames-Forces- Constraints-Dissipation Interface v0.1. The finite tooth-pair theorem is exact for a declared pitch-event model. Contact loads, wear, efficiency, lifetime, and tooth safety are not inferred from event counts.

Abstract

Two ideal gears with positive tooth counts N_1, N_2 admit an exact finite event model on

$$X_{N_1, N_2} = \mathbb{Z}_{N_1} \times \mathbb{Z}_{N_2}, \quad T_\sigma(i, j) = (i + 1, j + \sigma),$$

where $\sigma = -1$ denotes an external mesh and $\sigma = +1$ an internal mesh under the declared common angular orientation. Every orbit has length $\text{lcm}(N_1, N_2)$, the state space has $\text{gcd}(N_1, N_2)$ orbits, and one orbit covers every ordered tooth pair exactly when the counts are coprime. We separate this exact combinatorial layer from the active contact set, contact ratio, normal load, sliding, dissipation, and material wear. The signed speed ratio

$$\frac{\omega_2}{\omega_1} = \sigma \frac{N_1}{N_2}$$

identifies only the coprime count ratio, not the common scale that controls the cycle decomposition. For a planetary set, uniform subtraction of frame angular speed is formalized as an $(\mathbb{R}, +)$ -action; only a relative-observation protocol passes to its quotient. For a periodic non-circular kinematic model, we define the driver-period return map, its physical lift, invariant measures, and the Poincare translation number. No KAM conclusion is drawn. The result is a strict observation interface and a reproducible finite benchmark, not a new gear-design or tribology theory.

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1 Scope and observation protocol

The system descriptor is

$$d = (N_1, N_2, \sigma, \mathbf{g}, \mathbf{m}, \vartheta),$$

where $N_a \in \mathbb{N}_{>0}$, $\sigma \in \{-1, +1\}$, $\mathbf{g}$ records the declared geometry and frame conventions, $\mathbf{m}$ records the mesh class, and ϑ contains any constitutive parameters. The complete protocol is the typed chain

$$X_t \xrightarrow{\mathcal{R}_{\text{red}}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_t | Z_t) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t.$$

The record λ must state the tooth labels, event clock, reference frames, angular orientation, observation horizon, missed- event and misclassification policy, load calibration, sampling rates, noise model, and estimator.

Boundary 1.1 (Three layers that must not be identified). The pitch-event state x_k , the simultaneous active contact set $\mathcal{A}(t)$, and a material wear field are different objects. The first is finite combinatorics. The second requires tooth geometry and contact ratio. The third additionally requires loads, sliding, material data, lubrication, and a constitutive evolution law.

2 Fixed-axis circular kinematics

Let both angular coordinates be positive counterclockwise in one declared inertial frame. Put

$$\sigma = \begin{cases} -1, & \text{external mesh,} \\ +1, & \text{internal mesh.} \end{cases}$$

For an ideal constant-ratio pair, the phase constraint is

$$h(\theta_1, \theta_2) = N_1\theta_1 - \sigma N_2\theta_2 - \phi_0 = 0 \pmod{2\pi}. \quad (1)$$

On every smooth branch,

$$N_1\omega_1 - \sigma N_2\omega_2 = 0, \quad \rho_{2 \leftarrow 1} := \frac{\omega_2}{\omega_1} = \sigma \frac{N_1}{N_2}, \quad (2)$$

provided $\omega_1 \neq 0$. The direction in $\rho_{2 \leftarrow 1}$ is part of the symbol; writing only “gear ratio” is rejected.

Proposition 2.1 (Ideal mesh-reaction power). *Let $Q_c = \lambda dh$ be the reaction covector of the ideal constraint equation (1). Then its total generalized power vanishes on admissible velocities:*

$$\langle Q_c, (\omega_1, \omega_2) \rangle = \lambda(N_1\omega_1 - \sigma N_2\omega_2) = 0.$$

This statement concerns the ideal constraint reaction. Sliding friction, lubricant shear, impacts, backlash, bearing losses, and profile errors belong to separate force and dissipation terms.

3 Finite pitch-event dynamics

Definition 3.1 (Pitch-event state). Choose an event section that records one reference tooth-pitch crossing per advance of gear 1. The label state is

$$X_{N_1, N_2} := \mathbb{Z}_{N_1} \times \mathbb{Z}_{N_2}.$$

Under equation (2), its ideal event map is

$$T_\sigma(i, j) = (i + 1, j + \sigma) \pmod{(N_1, N_2)}. \quad (3)$$

Theorem 3.2 (Orbit length and cycle decomposition). *Every orbit of T_σ has length*

$$P = \text{lcm}(N_1, N_2).$$

The state space is the disjoint union of

$$g = \text{gcd}(N_1, N_2)$$

directed cycles of length P . A single orbit contains all $N_1 N_2$ ordered tooth pairs if and only if $g = 1$.

Proof. The period is the order of $(1, \sigma)$ in $\mathbb{Z}_{N_1} \times \mathbb{Z}_{N_2}$. It is the least positive k satisfying

$$k = 0 \pmod{N_1}, \quad \sigma k = 0 \pmod{N_2}.$$

Because $\sigma = \pm 1$, this is $\text{lcm}(N_1, N_2)$. Translation acts freely on each orbit, so every orbit has this size. The orbit count is therefore

$$\frac{N_1 N_2}{\text{lcm}(N_1, N_2)} = \text{gcd}(N_1, N_2).$$

□

Corollary 3.3 (Hunting-tooth criterion). *Every tooth of gear 1 is paired with every tooth of gear 2 on one event orbit exactly when $\text{gcd}(N_1, N_2) = 1$.*

The directed graph

$$G_{N_1, N_2, \sigma} = (X_{N_1, N_2}, \{(x, T_\sigma x) : x \in X_{N_1, N_2}\})$$

is consequently a union of equal directed cycles. Label translations $\ell_{a,b}(i, j) = (i + a, j + b)$ commute with T_σ , while $R(i, j) = (i, -j)$ conjugates T_σ to $T_{-\sigma}$. Thus P , g , and the cycle multiset are invariant under tooth relabeling and orientation reversal. The ordered label sequence is not invariant under those transformations.

Boundary 3.4 (Event-model boundary). When transverse or overlap contact ratio exceeds one, several tooth pairs may carry load simultaneously. The map T_σ then records a chosen pitch event, not the full active contact set. Undercutting, interference, flank modification, elastic deflection, and loss of contact are also absent from equation (3).

4 Ratio observation and identifiability

Let

$$\mathcal{O}_{\text{ratio}}(N_1, N_2, \sigma) = \sigma \frac{N_1}{N_2}.$$

This is a deterministic observation of the descriptor, not of an individual tooth-pair state.

Proposition 4.1 (Fiber of the magnitude-ratio observation). *Suppose*

$$|\mathcal{O}_{\text{ratio}}| = \frac{u}{v}, \quad \gcd(u, v) = 1.$$

Then every compatible positive tooth-count pair is uniquely

$$(N_1, N_2) = (cu, cv), \quad c \in \mathbb{N}_{>0}.$$

For this pair,

$$\gcd(N_1, N_2) = c, \quad \text{lcm}(N_1, N_2) = cuv.$$

Hence the speed-ratio magnitude does not identify contact-cycle count or return period.

Proof. The equality $N_1/N_2 = u/v$ gives $vN_1 = uN_2$. Coprimality implies $u \mid N_1$ and $v \mid N_2$, with one common positive multiplier c . The gcd and lcm formulas follow. $\square$

On a bounded design class

$$\mathcal{D}_M = \{(N_1, N_2, \sigma) : 1 \leq N_1, N_2 \leq M\},$$

the observation fiber is finite. On the unbounded class it is countably infinite. Therefore an information-loss statement must declare the descriptor domain; it cannot be inferred from a ratio alone.

Useful deterministic readouts include

$\mathcal{O}_{\text{pair}}(x_k) = (i_k, j_k)$	only with calibrated tooth labels,
$\mathcal{O}_{\text{phase}}(q) = (\theta_1, \theta_2) \pmod{2\pi}$	with declared encoders and frames,
$\mathcal{O}_{\text{ratio}}(d) = \rho_{2 \leftarrow 1}$	with signed orientation,
$\mathcal{O}_{\text{active}}(X_t) = \mathcal{A}(t)$	with contact-resolution policy,
$\mathcal{O}_{\text{load}}(X_t) = (N_\alpha, T_\alpha)_{\alpha \in \mathcal{A}(t)}$	with force calibration.

None of these channels is silently substituted for another.

5 Contact force, sliding, and dissipation

For an active candidate pair $\alpha = (i, j)$, let $g_\alpha(q) \geq 0$ mean separation and let $N_\alpha \geq 0$ be the compressive normal-force magnitude. In the smooth force regime,

$$g_\alpha \geq 0, \quad N_\alpha \geq 0, \quad N_\alpha g_\alpha = 0. \quad (4)$$

Let n_α point from gear 1 toward gear 2, and let $F_\alpha = N_\alpha n_\alpha + T_\alpha$ be the force on gear 2. The force on gear 1 is $-F_\alpha$. With

$$v_{\text{rel},\alpha} = v_{2,\alpha} - v_{1,\alpha}, \quad v_{T,\alpha} = (I - n_\alpha n_\alpha^\top) v_{\text{rel},\alpha},$$

a dissipative tangential law satisfies

$$T_\alpha \cdot v_{T,\alpha} \leq 0, \quad P_{\text{fric},\alpha} := -T_\alpha \cdot v_{T,\alpha} \geq 0. \quad (5)$$

If $J_{\text{rel},\alpha} \dot{q} = v_{\text{rel},\alpha}$, the generalized contact covector is

$$Q_\alpha = J_{\text{rel},\alpha}^\top F_\alpha \in T_q^* Q. \quad (6)$$

This pullback is required before contact forces are combined with other generalized forces.

In a smooth active-set interval, the nonnegative total frictional loss is

$$P_{\text{fric}} = - \sum_{\alpha \in \mathcal{A}(t)} T_{\alpha} \cdot v_{T,\alpha} \geq 0. \quad (7)$$

For a stationary autonomous model with ideal normal constraints, its energy balance has the interface form

$$\dot{E}_L = P_{\text{act}} - P_{\text{fric}} - P_{\text{other}}, \quad P_{\text{other}} \geq 0.$$

Moving supports, time-dependent tooth geometry, or imposed carrier motion require the explicit support-power or $-\partial_t L$ terms of the common mechanics interface.

Boundary 5.1 (Nonsmooth boundary). Contact onset, separation, impact, backlash switching, stick-slip transitions, and inconsistent rigid Coulomb contact require a nonsmooth inclusion or complementarity time-stepper. Existence, uniqueness, and numerical convergence are not proved here.

6 Counts, exposure, dissipation, and wear

For the pitch-event orbit $x_k = T_{\sigma}^k x_0$, define the count

$$C_{ij}(K) = \sum_{k=0}^{K-1} \mathbf{1}_{\{x_k=(i,j)\}}.$$

For a resolved continuous active set, define instead

$$I_{ij}(T) = \int_0^T \mathbf{1}_{\{(i,j) \in \mathcal{A}(t)\}} N_{ij}(t) dt, \quad (8)$$

$$D_{ij}(T) = \int_0^T \mathbf{1}_{\{(i,j) \in \mathcal{A}(t)\}} [-T_{ij}(t) \cdot v_{T,ij}(t)] dt. \quad (9)$$

Here C_{ij} is a count, I_{ij} has impulse dimension, and D_{ij} has energy dimension. They cannot be added or called equivalent.

For any declared nonnegative channel w_{α} with $W = \sum_{\alpha} w_{\alpha} > 0$, put

$$p_{\alpha} = \frac{w_{\alpha}}{W}, \quad H_b(p) = - \sum_{\alpha: p_{\alpha} > 0} p_{\alpha} \log_b p_{\alpha}, \quad b > 1.$$

If $W = 0$, the normalized distribution and its entropy are undefined.

Proposition 6.1 (Full-orbit event entropy). *For one full pitch-event orbit, the empirical distribution is uniform on $P = \text{lcm}(N_1, N_2)$ states. Therefore*

$$H_b = \log_b P.$$

Relative to the uniform distribution on all $N_1 N_2$ ordered pairs, the reachability support defect is

$$\Delta_b = \log_b(N_1 N_2) - \log_b P = \log_b \gcd(N_1, N_2). \quad (10)$$

For $N_1 N_2 > 1$, the normalized defect

$$\overline{\Delta} = \frac{\log_b \gcd(N_1, N_2)}{\log_b(N_1 N_2)}$$

is independent of b .

Neither C_{ij} , I_{ij} , D_{ij} , nor their entropies are a wear law. A wear prediction additionally needs a declared constitutive map with material hardness, local pressure, sliding distance, lubrication, temperature, surface evolution, and uncertainty.

7 Fixed-axis gear trains

Let a gear train have angular-velocity vector $\omega \in \mathbb{R}^m$. For every fixed-axis mesh edge $e = (a, b)$, use

$$N_a \omega_a - \sigma_e N_b \omega_b = 0.$$

For every rigid compound-shaft relation use $\omega_a - \omega_b = 0$. Stacking these rows gives

$$C\omega = 0. \quad (11)$$

The admissible ideal velocity space is $\ker C$; this formulation does not assume in advance that it is one-dimensional.

Along an oriented simple mesh path $v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_m$,

$$\frac{\omega_{v_m}}{\omega_{v_0}} = \prod_{r=1}^m \sigma_{e_r} \frac{N_{v_{r-1}}}{N_{v_r}}, \quad (12)$$

when all relevant velocities are nonzero. A closed ideal loop admits a nonzero compatible motion only if the corresponding signed product is one. Idler cancellation in a terminal ratio does not erase the idler's tooth-contact history, bearing loads, inertia, or losses.

8 Planetary frames as an explicit group action

Let $\omega = (\omega_s, \omega_r, \omega_c) \in \mathbb{R}^3$ denote sun, ring, and carrier angular velocities in one common axial orientation. A change to a frame rotating at angular speed α acts by

$$G_\alpha(\omega) = \omega - \alpha \mathbf{1}, \quad \mathbf{1} = (1, 1, 1). \quad (13)$$

It is an action of $(\mathbb{R}, +)$, because

$$G_\alpha \circ G_\beta = G_{\alpha+\beta}, \quad G_0 = \text{id}, \quad G_\alpha^{-1} = G_{-\alpha}.$$

Theorem 8.1 (Relative quotient and Willis relation). *The map*

$$\pi : \mathbb{R}^3 \longrightarrow \mathbb{R}^2, \quad \pi(\omega) = (\omega_s - \omega_c, \omega_r - \omega_c)$$

is constant exactly on the G_α -orbits and induces

$$\mathbb{R}^3 / \text{span}\{\mathbf{1}\} \cong \mathbb{R}^2.$$

For a simple ideal planetary set, the Willis relation is the quotient line

$$N_s(\omega_s - \omega_c) + N_r(\omega_r - \omega_c) = 0, \quad (14)$$

or, when the denominator is nonzero,

$$\frac{\omega_s - \omega_c}{\omega_r - \omega_c} = -\frac{N_r}{N_s}.$$

Proof. $\pi(\omega) = \pi(\omega')$ exactly when $\omega' - \omega = \alpha \mathbf{1}$ for some α . Thus its fibers are the action orbits and its kernel is $\text{span}\{\mathbf{1}\}$. Equation (14) depends only on $\pi(\omega)$. $\square$

Uniform frame subtraction is therefore a physical change of representation. It becomes an observational gauge redundancy only for a protocol of the form $\mathcal{O}_{\text{rel}} = f \circ \pi$, which deliberately discards common absolute spin. A laboratory encoder that measures ω_c separately does not pass to this quotient.

If the ring is fixed in the laboratory, $\omega_r = 0$, then

$$\omega_c = \frac{N_s}{N_s + N_r} \omega_s.$$

For $N_s = 30$, $N_r = 78$, and $\omega_s = 1 \text{ rad s}^{-1}$, this gives $\omega_c = 5/18 \text{ rad s}^{-1}$. For a standard equal-module sun-planet-ring geometry, the separate compatibility condition is $N_r = N_s + 2N_p$, giving $N_p = 24$ in this example.

9 Periodic non-circular kinematics

Let $r_1, r_2 : S^1 \rightarrow (0, \infty)$ be positive C^1 effective pitch radii, and assume the driver angle θ_1 is monotone. A kinematic rolling model is

$$\frac{d\theta_2}{d\theta_1} = \sigma \frac{r_1(\theta_1)}{r_2(\theta_2)}. \quad (15)$$

This equation is a model input. It does not prove manufacturability, absence of interference, force closure, or elastic stability.

For the lifted solution $\tilde{\theta}_2(\theta_1; x)$ with initial value x , define the one-driver-turn return lift

$$\tilde{F}(x) = \tilde{\theta}_2(2\pi; x), \quad \tilde{F}(x + 2\pi) = \tilde{F}(x) + 2\pi. \quad (16)$$

Uniqueness of the scalar C^1 flow makes the induced $F : S^1 \rightarrow S^1$ orientation-preserving, even when the output angle decreases along the driven orbit.

Theorem 9.1 (Translation number with declared lift). *For the orientation-preserving circle homeomorphism F and the physical lift $\tilde{F}$, the limit*

$$\tau(\tilde{F}) = \lim_{n \rightarrow \infty} \frac{\tilde{F}^n(x) - x}{2\pi n} \quad (17)$$

exists and is independent of x . Replacing the lift by $\tilde{F} + 2\pi k$ changes τ by k ; hence $\rho(F) = \tau(\tilde{F}) \bmod \mathbb{Z}$ is lift-independent. If μ is any F -invariant Borel probability measure, then

$$\tau(\tilde{F}) = \frac{1}{2\pi} \int_{S^1} (\tilde{F}(\tilde{x}) - \tilde{x}) d\mu(x),$$

where the displacement is the well-defined periodic function induced by the selected lift.

For constant radii, $\tilde{F}(x) = x + 2\pi\sigma r_1/r_2$, so $\tau = \sigma r_1/r_2$. Elasticity, backlash, impact, or controller hysteresis may destroy the single-valued homeomorphism hypothesis. No KAM persistence, Diophantine stability, or locking theorem follows from equation (15) alone.

10 Reproducible benchmarks and machine gates

(N_1, N_2)	gcd	lcm	orbit fraction	$ \rho_{2 \leftarrow 1} $
(12, 25)	1	300	1	12/25
(12, 18)	6	36	1/6	2/3
(2, 3)	1	6	1	2/3
(4, 6)	2	12	1/2	2/3

The release tests verify the orbit theorem by exhaustive enumeration over a finite tooth-count range, ratio fibers, graph conjugacy, explicit entropy bases, contact complementarity, dissipation signs, force-power covariance, train path products, the planetary group action and quotient, the Willis example, and translation-number convergence for benchmark circle maps.

Gate	Required declaration	Rejected inference
Event semantics	pitch section and label convention	event pair equals simultaneous contact set
Ratio direction	ω_2/ω_1 , sign, and frame	unnamed “gear ratio”
Frame action	G_α and observed quotient	calling every carrier subtraction a gauge
Contact force	gap sign, active set, force-on-body convention	unsigned or double-counted reactions
Dissipation	$P_{\text{diss}} \geq 0$ with relative velocity	event count used as energy loss
Exposure	count, impulse, and energy kept dimensionally distinct	load count called wear
Entropy	base, support, positive-total guard	log without base or zero normalization
Rotation number	return map, lift, and hypotheses	KAM language from a local ratio equation

11 Claim boundary

Statement	Status	Required hypotheses
Finite orbit decomposition	theorem	positive integer tooth counts; pitch-event map equation (3)
Ratio fiber	proposition	exact constant ratio; positive tooth counts
Ideal reaction power	proposition	smooth ideal holonomic constraint
Willis quotient	theorem	common axis convention; ideal simple planetary kinematics
Translation number	classical theorem	orientation-preserving circle homeomorphism and selected lift
Contact friction	constitutive model	declared active set, force convention, smooth force regime
Wear prediction	absent	requires an external validated material-evolution model

This revision establishes no new gear-design standard, load-capacity formula, lubrication model, lifetime law, or safety result. Its exact mathematics is elementary finite-group dynamics plus classical circle rotation theory. The contribution is a strict separation of descriptor, event, contact, frame, dissipation, and observation layers suitable for GO Core validation.

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